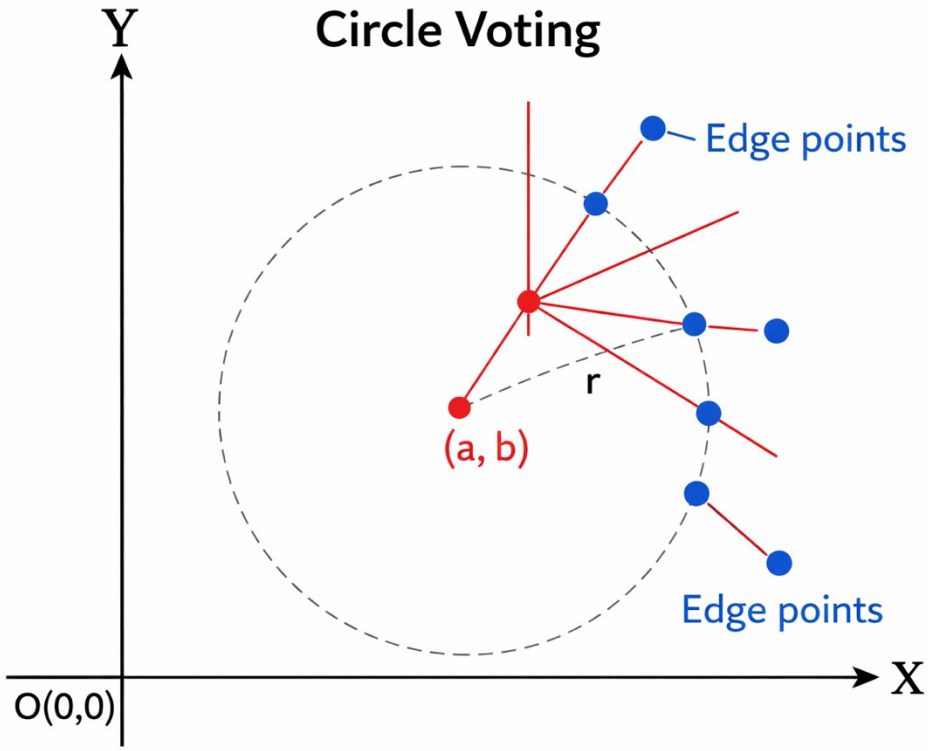
# Unit 3 - Session 4

## SLO 1: Extend HT to detect circles in images.

Duration: 10 Minutes Activity

**Question:**




**Q1.** What do the blue points in the diagram represent?

Answer: The blue points represent the edge points of the circle.

**Q2.** What does the red point labeled $(a, b)$ indicate?

Answer: The red point labeled (a, b) represents the center of the circle.

**Q3.** Why are multiple red lines drawn from the edge points toward the center?

Answer: Each edge point votes for possible circle centers along the radial direction.

**Q4.** What geometric property ensures that all voting lines intersect at the same point?

Answer: All radii of a circle pass through the same center point.

**Q5.** What does the dashed circle in the diagram represent?

Answer: The dashed circle represents the circle being detected, with center (a, b) and radius r.

**Q6.** How is the radius *r* of the circle determined once the center is known?

Answer: The radius is the distance from the known center to any point on the circle boundary.

**Q7.** In the Circular Hough Transform, what parameters are stored in the accumulator?

Answer: The accumulator stores the circle parameters (a, b, r): center coordinates and radius.

**Q8.** What does a peak in the accumulator space signify in circle detection?

Answer: A peak signifies a circle parameter set receiving many votes, indicating a detected circle.